

Integral of Complex Exponentials

Timothy Tso

October 6, 2026

The problem we are trying to solve is the integral

$$\int_{-\infty}^{\infty} [e^{ikx}] dk = 2\pi\delta(x) \quad (1)$$

where we put brackets around the complex exponential to emphasize that the term is understood to be a distribution, not a function. Notice that we cannot take this integral: for any point not equal to zero the complex exponentials oscillate and never reach a limit point.

Instead, what the above equation is saying is that the two sides are equal when convolved with a certain test function $f(x)$. This means that

$$\int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} [e^{ikx}] dk \right) f(x) dx = \int_{-\infty}^{\infty} 2\pi\delta(x) f(x) dx \quad (2)$$

We can swap the order of the integrals on the side of the above equation to evaluate the expression. Notice that, unlike the original integral, this integral is possible to evaluate. We can think of the integral swapping as compressing the failure to integrate into a single value that is integrable, and evaluate the final result.